Age (in years)	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
72	4412	-	-	-	-	-	-
73	3724	-	-	-	-	-	-
74	3201	642	-	-	-	26567	-

Solution:

Values of the missing entries can be calculated from the given data applying the respective formulae as follows:

The number of persons who die before reaching age $x = 72$ and 73 years can be calculated as

$$d(72) = l(72) - l(73)$$

$$= 4412 - 3724$$

$$= 688$$

$$d(73) = l(73) - l(74)$$

$$= 3724 - 3201$$

$$d(73) = 523.$$

Values of $q(x)$ are estimated as

$$q(72) = \frac{d(72)}{l(72)}$$

$$= \frac{688}{4412}$$

$$= 0.1559$$

$$q(73) = \frac{d(73)}{l(73)}$$

$$= \frac{523}{3724}$$

$$q(73) = 0.1404$$

$$q(74) = \frac{d(74)}{l(74)}$$

$$= \frac{642}{3201}$$

$$= 0.2006.$$

Values of $p(x)$ are estimated from the corresponding values of $q(x)$ as

$$p(72) = 1 - q(72)$$

$$= 1 - 0.1559 = 0.8441$$

12th_Statistics_EM_Unit_8.indd 222 1/28/2020 6:38:12 PM